MicroPicoDrive NG firmware manual

This manual shows you how to prepare a microSD card, work with cartridge images on the Sinclair QL, and set everything up in System Tools. It matches firmware version 2.10.1.

Fitting the board inside the QL — including the Pico and the optional vibration motor — is covered by the hardware installation manual in the hardware repository.

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The two parts and firmware versions

The **mainboard** stays inside the QL and connects to its Microdrive bus. It carries the Raspberry Pi Pico that runs the firmware. The removable **cartridge board** carries the display, four buttons and microSD socket. It plugs into the mainboard's edge connector.

A **cartridge image** is a file on the microSD card that behaves like one QL Microdrive cartridge. When you *mount* an image, the QL sees it as a cartridge sitting in the drive; when you *eject* it, you are back in the file browser. Mounting and ejecting happen on the screen — they are not the same thing as physically plugging in or pulling out the cartridge board.

Feature	Lite: Raspberry Pi Pico / RP2040	Full: Raspberry Pi Pico 2 W / RP2350
Image browsing, QL loading and saving	Yes	Yes
Display and motor settings	Yes	Yes
SD card firmware updates	UF2 file	BIN file and matching JSON manifest
Bluetooth file management and updates	No	Yes
Revert Firmware	No	When another firmware slot is available
BOOTSEL Mode menu entry	Yes	No

Install the firmware that matches the Pico you actually fitted. The Pico 2 W's wireless features use Bluetooth Low Energy (BLE), not Wi-Fi.

Preparing the microSD card

Format and copy files

1. Back up any existing files before formatting: formatting erases them.
2. Use a card reader on your computer. Prepare a single normal data volume in **FAT32 or exFAT**. Both are enabled in this firmware; FAT12 and FAT16 are also supported by its filesystem library. NTFS and APFS are not supported. Check the selected device carefully before applying the format.
3. Copy valid `.mdv` cartridge images onto the card. You can put them in the root or organise them into folders. Extract downloaded ZIP archives on the computer first.
4. Safely eject the card from the computer and insert it into the cartridge board's microSD socket in the socket's indicated orientation. Do not force it.

You don't need any firmware files or a `CONFIG.CFG` on the card just to browse images. We haven't tested every card size, and a supported filesystem alone doesn't guarantee a particular card will behave — so run **System Tools > SD Check** once on a freshly prepared card.

Example card layout:

```
/
  Games/
    CHESS.mdv
    CHESS.mdv.thumb       optional matching artwork
  Work/
    NOTES.mdv
  CONFIG.CFG              created when you tag an auto-load image
  .update/                needed only for firmware updates
```

The QL never sees this folder tree — it only sees the inside of the mounted cartridge image. Copying a loose QL program or a ZIP file onto the card does not put it inside a cartridge; use an image authoring tool on your computer for that. The device itself cannot create new blank images. This project's companion app for authoring `.mdv` images is Sinclair MDV Builder — see the addendum at the end of this manual for a quick reference.

Image and filename limits

Item	Requirement or behaviour
MDV image	174,930 bytes
Browser file filter	`.mdv`, case-insensitive
Auto-load extension	Use all-lowercase or all-uppercase: `.mdv` or `.MDV`; mixed-case extensions do not auto-load in this version
File and folder names	Keep each to 63 bytes or fewer, including extension; longer names are truncated by the browser and may fail to open
Full path	Keep below 300 bytes, including separators; shorter paths also avoid thumbnail lookup limits
Entries per folder	At most 64 eligible files and folders; use subfolders for larger collections
Hidden entries	Hidden/system files and names starting with a dot are omitted

Use short, simple filenames when preparing your first card. An image with the wrong size reports **Bad format**. Renaming an unrelated file to `.mdv` does not convert it into a cartridge image.

When a folder exceeds the entry limit, a non-selectable `...` row appears. The firmware collects entries in on-card order before sorting them for display, so the omitted files are not necessarily the last alphabetically.

Optional artwork lives beside the image. The preferred `CHESS.mdv.thumb` format is a raw 160 by 80 pixel RGB565 big-endian image, exactly 25,600 bytes. The firmware also accepts legacy `CHESS.thumb` artwork as uncompressed 24-bit BMP data. A renamed JPEG or PNG will not work. Missing artwork does not prevent the cartridge from loading.

First start and everyday use

First-start check

1. Start with a card containing at least one known-good image and no firmware update files. Insert the cartridge board and power on the QL.
2. Wait for the display to initialise and the file browser to appear. Without a readable card, the firmware shows a waiting screen instead.
3. Use K1/K2 to highlight an image and press K3 to mount it. Wait for loading to finish and the cartridge screen to appear. (See Buttons below for the full button map.)
4. At the QL's SuperBASIC prompt, list the corresponding drive. For a unit installed as drive 1:

```
DIR mdv1_
```

Use `DIR mdv2_` if it replaces drive 2. A directory listing confirms the QL can read the mounted image. If it fails, check the image and drive position before investigating the physical installation with power disconnected.

Before you save any real work, read **Loading programs and saving your work** below — saving on the QL and saving to the SD card are two separate steps.

Buttons

The button labels are K1 through K4; a long press is approximately one second.

Button	File browser	Mounted cartridge screen	System Tools / reports
K1 / UP	Previous entry	Eject image	Previous item / scroll up
K2 / DOWN	Next entry	Eject image	Next item / scroll down
K3 / SELECT	Open folder or mount image	Save image to SD	Activate item or cycle its value; close a report
K4 / TOOLS	Open System Tools	Shows `Eject first`	Leave menu or close report
Hold K3	Tag or untag image for auto-load	No special save action	On `Pair Device`, request forgetting all pairings
Hold K4	Pico 2 W: shortcut to Connect	Eject first	Use the displayed mode's controls

Hold K4 while the drive powers up and it switches the display Panel type — a rescue for a screen you cannot read. See **Panel** under UI Options.

Folders appear in brackets. Select `[..]` to go up one folder. During QL drive activity the normal cartridge buttons are ignored; wait for the QL operation to finish. The activity LED indicates drive selection/activity and also blinks during SD operations. It does not distinguish QL reads from writes.

In Yes/No dialogs, K1 or K2 changes the answer and K3 confirms it. Confirmation dialogs time out after about 30 seconds without carrying out the requested action. Firmware update timeout handling is explained below.

Loading programs and saving your work

Once mounted, use the image like a Microdrive cartridge. For example, `LOAD mdv1_program` loads a BASIC program named `program`; some application cartridges use `LRUN mdv1_boot`. Follow the program's own loading instructions and substitute the correct drive number.

Saving on the QL and saving to the SD card are two separate steps. QL writes change the mounted image in RAM. They are not automatically written back to the card.

1. Save your work from the QL application, or use a command such as `SAVE mdv1_test` for a BASIC program.
2. Wait until the QL has finished using the drive.
3. Press **K3 on the cartridge screen**. Wait for **Saved** before removing anything or turning off power. This overwrites the original image file.
4. Press K1 or K2 to eject the image when you want to choose another one.

If you eject after QL writes without saving to SD, the device asks **Eject?** / **Unsaved changes**. **Yes discards those changes**; it does not save them. Choose No, press K3 to save, then eject again to keep your work.

Keep backups on your computer. Saving rewrites the image in place, so a failed or interrupted SD save can leave the on-card file incomplete. If you see **Save failed** / **Retry save!**, keep the unit powered and the image mounted, resolve the card problem and retry K3. The firmware retains the RAM image after a failed save, but ejecting or losing power loses that recovery opportunity.

Auto-load a favourite image

In the browser, highlight an image and hold K3 for about one second. A red marker in the left gutter identifies the tagged image. Only one image per card can be tagged; tagging another replaces the previous choice. Hold K3 on the tagged image again to clear the tag.

The firmware creates or updates `CONFIG.CFG` in the card root, recording the image's path in a `FILE=` entry. It tries that image when the card is mounted during startup. Tagging does not run a QL program; it only mounts the cartridge. If you rename or move the image, tag it again at its new location. Use the buttons to manage this file rather than editing it by hand.

Removing a card or cartridge board

Save to SD, eject the image, and wait for any file operation to finish before removing the SD card or cartridge board. The cartridge board supports powered insertion/removal, but this does not preserve unsaved work or make interruption of an SD write safe. Never remove it during a firmware update.

If the SD card was swapped while an image remained mounted, saving may report **Different card** / **Save blocked**. Reinsert the original card and retry K3; the RAM image is retained. This protection compares volume serial numbers, so it is not a substitute for the save-and-eject procedure.

System Tools reference

Eject the cartridge image, then press K4 in the browser or an available waiting screen. K1/K2 moves through the list, K3 activates an item or cycles its value, and K4 or **Exit** leaves the menu. **UI Options** and **Motor** are section labels, not actions.

Display and motor changes take effect in the session and are saved to onboard flash when you leave System Tools. Exit the menu before powering off. These settings stay with the mainboard when you change SD cards;

the auto-load tag stays on each card in `CONFIG.CFG`.

Diagnostics and maintenance

Menu item	Availability	What it does
SD Check	Both builds; only if a card mounted when Tools opened	Reports filesystem, total/free space, and write/read speeds using a 32 KB test file with sampled data verification
System Info	Both	Shows firmware version, CPU type, clock, chip temperature, RAM used/free, flash capacity and application size
LED & Motor Test	Both	Blinks the activity LED and runs the fitted motor for five seconds; intentionally bypasses the Motor on/off setting
BOOTSEL Mode	Lite only	After confirmation, reboots into USB programming mode; normal drive operation stops
Exit	Both	Leaves Tools and saves changed display/motor settings

SD Check writes to the card. It creates or overwrites `UIEXTTST.BIN` in the root and deletes it when the read/check sequence completes. Do not keep your own file under that name. A failed test can leave it behind. The test is a quick access check, not a full-card surface test or filesystem repair.

Use K1/K2 to scroll diagnostic reports and K3/K4 to return. If you inserted a card while Tools was already open, exit and reopen Tools to make SD Check available. System Info temperature is the Pico's sensor reading, not a measurement of the whole QL.

UI Options

Press K3 repeatedly to cycle a setting. The label shows the current value.

Setting	Values	Default without saved settings	Effect
Caption	Top, Mid, Bottom	Bottom	Positions the caption on the cartridge artwork screen
Dark Mode / Light Mode	Dark or Light	Dark	Switches display background/text and selection colours
Path Bar	On, Off	Off	Shows or hides the red path title bar in the file browser
Rainbow	Top Left, Top Right, Btm Left, Btm Right, Off	Top Right	Positions or hides the Sinclair rainbow corner decoration
Panel	Type 1, Type 2	Type 2	Matches the setting to the display fitted to your cartridge

About Panel. Two versions of the 0.96" display module are sold, and they need slightly different settings. If your screen shows the wrong colours — red looking light blue, white looking black — or a thin line of speckled rubbish down one or two edges, switch this to the other type. The picture corrects itself immediately. Everything else is unaffected, so if the screen already looks right, leave it alone.

There is no way for the drive to detect which display you have, which is why you pick it here.

You do not have to find this menu to change it, which matters when the screen is the thing that is hard to read. **Hold K4 while the drive powers up**: the Panel type switches before the display is even started, so it comes up in the other setting, and a buzz confirms it. The choice is saved straight away, so you only do it once.

New drives are set to **Type 2**, the version currently being sold. If you have an older drive that has been

working fine, an update leaves it on Type 1 and nothing changes for you.

Motor

These options only do something when the optional vibration motor is fitted. They never affect the emulation or your saved cartridge data.

Setting	Values / default	Effect
Motor	On / Off; default On	Master switch for normal vibration feedback
Cart	On / Off; default On	Feedback for physical cartridge-board insertion/removal detection
Transfer	On / Off; default On	Completion feedback for supported Bluetooth transfer/update operations
Alerts	On / Off; default On	Double-buzz feedback for error/alert events
Long Press	On / Off; default On	Feedback when a button reaches its long-press threshold
Load/Save	Off, 250ms, 500ms, 1s; default 500ms	Runs the motor during SD image load/save and for the selected extra time afterwards

The `Load/Save` time is how long the motor keeps running *after* the operation finishes — a little spin-down, like a real drive. The motor always runs for the whole load or save itself. Turning **Motor** off silences all feedback at once while remembering your individual settings, and **LED & Motor Test** still runs the motor regardless, so you can always check the hardware.

Bluetooth and firmware slots: Pico 2 W only

Menu item	What it does
Connect	Opens a BLE session for a previously paired client to manage SD files and query the device
Pair Device	Allows a new client to pair; use a short K3 press
Paired Devices	Lists saved pairings; select one with K3 to open `Forget Device?`, or K4 to return
Update Firmware	Opens the wireless firmware update mode
Revert Firmware	Appears when the other slot contains firmware; asks before switching back and rebooting

The radio only runs while you are in one of these menu modes, and only with no image mounted and the QL drive idle — during normal cartridge use it is completely off. K3 or K4 cancels the connect, pairing and update waiting screens; if a firmware installation is already running, let it finish.

Holding K3 on **Pair Device** requests **Forget all paired devices?**. Confirm only if you want to remove every stored pairing. After forgetting a device, remove its old pairing at the client too before pairing again.

Revert Firmware invalidates the current slot and boots the other one. It is not a permanent toggle between two retained versions; reinstall the newer firmware if you want to return to it later.

Firmware updates

Use a release intended for your board. Save and eject your mounted image first, and keep power and the cartridge board connected throughout installation.

From the microSD card

1. On your computer, create a folder called `.update` at the card root.
2. Copy the release's SD-update files into it:

Build	Files
Lite / RP2040	The release's update `.uf2`
Pico 2 W / RP2350	The `.bin` and its matching `.json` manifest

3. Safely eject the card, fit it to the device and power on. The update check happens before image auto-load. `.update` is hidden from normal browsing.
4. Review the installed and offered versions, choose Yes or No with K1/K2, and confirm with K3.

Yes installs and reboots. **No deletes the offered update files**, preventing another prompt. Waiting for the 30-second timeout leaves the files for another offer at the next startup. A differing older version can be installed from SD; the prompt marks it as older. Pico 2 W ignores a same-version SD update. Lite skips an image whose contents already match the running application; a changed image with the same version number can still be offered.

Over Bluetooth

The usual way to update a Pico 2 W wirelessly is the **MicroPicoDrive Companion** phone app: it finds the latest release, downloads and checks it, and sends it to the unit. Pair the phone once, eject the mounted image, and start the update from the app — on firmware 2.7.0 or later the app switches the unit into Update Firmware mode by itself; on older firmware, choose **System Tools > Update Firmware** on the unit first. Any client implementing this project's BLE update protocol can do the same; this repository does not include a dedicated firmware upload client.

Quick reference: the MicroPicoDrive Companion app

When you want to...	Start in the app at
Connect your phone to the unit	Home > Connect to a unit
Update the unit's firmware	Home > Firmware
Send a cartridge to the unit	Local MDV Catalog > long-press a cartridge
Retrieve a cartridge from the unit	Remote MDVs > the retrieve button
Back up the whole unit into a folder	Remote MDVs > Back up
Create a blank cartridge on the SD card	Remote MDVs > Create empty
Reboot, identify or rename the unit	Home > Pico Tools

Two things the app deliberately does not do: copying is not synchronizing — changes made on the QL do not update the phone's copy, so retrieve the cartridge again when you want the newer contents — and sending a cartridge does not mount it; load it with the unit's own buttons afterwards.

The wireless route requires a strictly newer version. Confirm installation when prompted. No discards the upload; if the prompt times out, the firmware attempts to store the verified update in `.update` on the SD card for the next boot. That deferred route requires a working writable card. Watch for any reported storage error rather than assuming the update was saved.

Initial programming and USB recovery

Initial programming differs from an ordinary SD update. For Lite, the merged `MicroPicoDrive-full.uf2` includes the bootloader needed on a blank Pico. The application UF2 alone is not a complete first installation.

Hold the Pico's BOOTSEL button while connecting USB to enter its `RPI-RP2` programming drive, then copy the merged image. The Lite menu's **BOOTSEL Mode** also enters this mode on an already running unit.

Pico 2 W first installation additionally needs its partition table and radio firmware, followed by the sealed application image. Follow the release's board-specific instructions and the repository's firmware and build documentation. Confirm the hardware's USB power arrangement before servicing an installed mainboard.

Troubleshooting

Symptom or message	What to check or do
No display	Check cartridge-board seating and QL power. Disconnect power before inspecting internal ribbon or Pico connections
Waiting for SD / card will not mount	Check seating and a supported filesystem; try a backed-up, freshly prepared card
No images listed	Extract archives, check the `.mdv` extension and hidden attributes, and open the correct folder
A file is missing / `...` at the end	Split a large folder into smaller ones; keep names within the browser limits
Bad format	Check image format and exact file size; recover a clean copy if necessary
Load failed	Check the original filename/path and card readability; a truncated long name can cause this
QL cannot read the image	Wait for mounting, use the correct `mdv` number and try a known-good image; inspect wiring only with power disconnected
Buttons do not respond	Wait for QL drive activity to finish; normal cartridge controls are gated while selected
Eject first	Save with K3, eject with K1/K2, then open Tools
Different card / Save blocked	Reinsert the original card and retry K3 without ejecting the image
Save failed / Retry save!	Keep power on and the image mounted, resolve card access/free-space problems and retry; the on-card image may already be incomplete
Event lost	The RAM image is untrusted. Only K1 eject works; saving is blocked. Reload a known-good saved image
Colours look wrong, or a speckled line down an edge	Hold K4 while powering up to switch the display Panel type, or change **Panel** in UI Options
Settings did not persist	Leave Tools through Exit or K4 before power-off; settings cannot be saved while the drive is in use
SD Check absent	Fit a readable card, leave Tools and reopen it
No vibration	Check Motor and the individual event settings; try LED & Motor Test, then inspect optional hardware with power disconnected
BLE options absent	Confirm Pico 2 W firmware and eject the mounted image; Lite has no Bluetooth
Pairing fails after forgetting a client	Remove the stale pairing on the client, then pair again through Pair Device

One known quirk: a mainboard powered up on the bench, outside a QL, can pick up a false drive-select signal and leave the activity LED stuck on — which also blocks file operations and settings saves. Power-cycle the board and try again. Inside a QL this doesn't happen; it is purely a bench condition.

Addendum: creating images with Sinclair MDV Builder

Sinclair MDV Builder is this project's free Windows companion app for creating, inspecting, editing and extracting files from `.mdv` cartridge images. It needs the .NET 8 Desktop Runtime. The basics at a glance:

What you want	Where it is
A new, empty cartridge	New empty cartridge — creates a valid 174,930-byte image
A cartridge from files you have	New cartridge from loose files, or drag and drop them in
A cartridge from a downloaded ZIP	New cartridge from a ZIP archive
Add files to an existing image	Open the `.mdv`, then Import file(s) or Import from ZIP
Get files out of an image	Extract a file, Extract All, or Export to ZIP
Look inside an image	Directory listing, per-file hex/text viewer, sector map, media info
Housekeeping	Duplicate, Rename, Delete, and Set Exec on any file

Points that matter for this device:

- QL file type and data-space survive ZIP export/import (qlzip-compatible),

so executables keep their headers.

- Save the finished `.mdv`, then copy it to the SD card as described in

Preparing the microSD card.

- Its images round-trip the builder's own loader and tests; validation on a

stock QL ROM is still in progress, so keep a backup of anything important.

Addendum: the Bluetooth protocol for developers

On the Pico 2 W, the Connect mode in System Tools speaks a documented Bluetooth Low Energy protocol, so you can write your own tool — an app or a script — to manage the SD card wirelessly. The full wire protocol lives in BLE_PROTOCOL.md in this repository. What it lets a client do, at a glance:

Area	Operations
Files	list folders, upload and download images (SHA-256 verified), rename, delete
Cartridge labels	read and rewrite the medium name inside an `.mdv` image
Device	info and health snapshots, identify (flash the screen), set the device name
Firmware	hand the device over to its wireless update mode

The essentials: pair once per computer while the device shows **Pair Device** (a 6-digit passkey appears on the device display), then talk to it whenever it sits on the **Connect** screen. Uploads are refused while a cartridge image is mounted, and every transfer is checksummed — a committed upload is a verified upload. The repository also ships `tools/mdvtool.py`, a working command line client that doubles as example code.

Documentation basis

Operation and menu details were checked against UserInterface.c, UserInterfaceExtension.c, SD Check, System Info, filesystem configuration, and BLE implementation.